

Human-Centered Design:
From "bare minimum" to "emotionally
supportive" lactation rooms
nurturemode

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1 Abstract

Lactating parents who are returning to work face significant structural and psychological problems; this often leads to premature cessation of breastfeeding due to the lack of supportive environments [1]. This project aims to address these issues through a human-centred design lens; the project’s overall goal is to design an ambient personalisation system for workplace lactation rooms. Through iterative research with proxy users, the study explored how sensory modulation, specifically lighting, sound, and visual presets, can facilitate a psychological transition from a ”worker mindset” to a ”nurturing mode”. The results of this study indicate that environmental autonomy significantly enhances perceived comfort and dignity, though technical and ecological limitations remain. This work contributes to the discourse on design justice [2] by advocating for socio-technical interventions that prioritise the well-being and autonomy of lactating employees in the professional sphere.

2 Introduction

For many parents, breastfeeding and pumping at work are everyday realities, yet they remain difficult to manage within most professional environments. Balancing the physical demands of milk expression with the emotional weight of doing so within a working day can place a considerable burden on lactating parents. Although Dutch law requires employers to provide time and an appropriate space for lactation, in practice these provisions are often treated as a compliance task and reduced to the bare minimum. As a result, parents are left to navigate stress, lack of privacy, emotional discomfort, and inconsistent or unclear organisational support.

This project addresses this gap between legislation and lived experience through a human-centred design lens. The study first aims to understand how lactation at work is experienced and where friction emerges, in order to define a clear scope and objective for design. To do so, qualitative data is gathered through a series of semi-structured interviews. The interview material is then analysed through affinity mapping and the labelling of recurring themes, allowing the project to translate subjective experiences into concrete design directions and a coherent design vision.

Based on this understanding, the project proceeds through iterative ideation, prototyping, and evaluation. The overall goal is to develop and deploy an artifact that better supports lactating parents in workplace lactation rooms, while also examining the interactional and experiential effects that arise when a digital system is introduced into an intimate and emotionally sensitive context.

3 Understanding

3.1 Challenge

Context. Currently, one of the most prominent societal priorities is promoting inclusiveness and equal opportunities for every individual, regardless of their gender, race or sexual orientation. University of Twente, through its Shaping2030 strategy, aims to bridge the gap between biases and aims to create a diverse, equitable, and inclusive academic community. Within this wider mission, this project focuses on the experiences of lactating people returning to work after childbirth. Although breastfeeding provides important health benefits for both child and parent, Dutch breastfeeding rates remain below global recommendations: only 19% of infants are exclusively breastfed at six months, and 28% receive any breastfeeding during that period [3]. Many parents stop earlier than intended due to the difficulty of combining breastfeeding with professional responsibilities. As studies show, only 32% of Dutch breastfeeding mothers have access to any lactation room, from which only 24% have the possibility to lock the room from the inside for privacy [4]. This makes workplace support for lactation not only a health issue but also a matter of equity and inclusion.

Under the Dutch Law, employers are required to provide the opportunity to express milk or breastfeed at the workplace for up to 9 months after childbirth. This entails both the time and space for this activity, and they are also required to provide a dedicated lactation room that is clean, quiet, and lockable from the inside, equipped with a bed or couch for resting. This period of time is considered paid working hours, and in the event of impossibility to create such arrangements at the location, employees must be allowed to express milk or breastfeed at home, with travel time included as working hours. But, in reality, often lactating parents encounter issues with this, and this gap between legislation and lived experience highlights that workplace support for lactation is not only a health and comfort issue but also one of equity, dignity, and inclusion.

Problem Statement. A substantial gap exists between lactating employees' practical and emotional necessities and the facilities or organisational practices designed to help them. Although many workplaces exceed the minimum functional requirements, these spaces typically fail to accommodate crucial psychological, environmental, and dignity-related variables such as privacy, comfort, predictability, and stigma reduction. This misalignment can lead to increased stress, decreased general well-being, and early termination of breastfeeding, especially for people who work in less flexible or supportive environments.

Aim. The goal of this research is to explore how lactating parents experience breastfeeding while working, identify the challenges they experience, and design solutions for the breastfeeding spaces that better meet their physical, emotional, and logistical needs. This includes incorporating ideas of diversity, equity, inclusion, and justice to guarantee that the final design serves a wide range of people.

Particular Dynamic. This project focuses on how privacy, emotional comfort and accessibility can be integrated into the designated spaces for expressing milk at the workplace. Early findings show that lactating parents often experience stress related to the availability of suitable spaces, fear of being interrupted, and the pressure to remain discreet. The emotional and logistical issues are the main elements that shape the breastfeeding experience. Understanding how these factors influence daily routines is essential in designing solutions that support both the practical and psychological needs of lactating employees.

Stakeholders. The primary stakeholders are lactating parents who return to work and must manage breastfeeding routines throughout the workday. Secondary stakeholders include direct employers, who determine workplace policies and schedules; building managers, who oversee the allocation and maintenance of suitable rooms; work colleagues, whose attitudes and actions influence the social environment; and support healthcare professionals, who provide prior guidance. Family members and partners also play an important role by offering emotional and logistical support outside the workplace.

Positionality. As a design team, the understanding of this challenge is influenced by our own identities and life experiences. Most of the team members are young, international students, many of whom are male, have no siblings, and have no first-hand experience with pregnancy, childbirth, or breastfeeding. Acknowledging this gap requires the team to engage with empathy, remain mindful of potential biases, and centre the lived experiences and voices of lactating parents throughout the design process. The main goal as a design team is to ensure that the design process is anchored by acknowledging personal positional limits.

3.2 Interviews

3.2.1 Scope

The main goal of the interviews conducted was to collect primary data that would move beyond the "structural domain" of laws and written policies to understand the "lived experience" of lactating parents returning to work [5]. Although the initial problem statement established that legal frameworks exist in the Netherlands, statistical data

suggests that these rights are often inaccessible in practice. As a result, interviews aimed to collect qualitative "missing data" regarding the invisible hurdles, social stigma, and power dynamics that prevent employees from exercising their rights [5].

Drawing on the Social Ecological Model and the specific challenges identified in Dutch universities, the inquiry focused on key dimensions of the lactation experience [6]:

- **The Physical Environment:** It was important to know if the current lactation facilities meet more than just the minimum legal requirements (a lock and a chair). The provision of a restorative, hygienic, and psychologically safe environment that stimulates the let-down reflex, was investigated [6].
- **Work Culture and Organisation:** To understand how the 25 per cent lactation time entitlement interacts with rigid academic schedules and work routines it was asked if workload expectations are adjusted to account for this time, or if lactating parents are forced to work overtime to compensate, creating a "double burden" [6].
- **Policy Awareness:** Determining if there is an information gap was key. Do employees know their specific rights upon returning to work, or does the university fail to proactively communicate this information, leaving the burden of research on the parent? [6].
- **General Pain:** Hentges and Pilot (2021) [6] explicitly mention that breastfeeding is an "extra job" that takes immense energy and that physical issues like pain or babies refusing bottles are major stressors for returning mothers.
- **Tools and Logistics:** The interviews investigated the design and logistical issues associated with lactation tools. The aim was to identify the challenges mothers faced with equipment such as breast pumps and bottles and the subsequent difficulties related to milk storage (e.g., shared office fridges), treating them as potential physical or psychological barriers within the workplace.

The interviews were tied directly to the problem statement, which highlights the gap between the equality promised by Dutch law (everyone has the right to pump) and the lack of equity in practice (not everyone can pump without professional or social penalty). By gathering these insights, the scope is to validate that the problem is not merely a logistical issue of "missing rooms" but a deeper "justice issue" related to how the university values care work [2, 5]. The insights gathered here are used to reframe the design challenge from simply providing infrastructure to designing a supportive system that prioritises the well-being and dignity of the lactating parent.

3.2.2 Method and protocol

To gain a deep understanding of the "lived experience" of lactating parents, semi-structured interviews were utilised. This qualitative method was chosen because it allows for a flexible inquiry that can adapt to the interviewee's responses while ensuring that all key themes are covered. In line with the principles of Design Justice, these interviews were approached using a "participatory" mindset, viewing the participants not as test subjects but as experts in their own experience [2]. To move beyond surface-level answers and uncover the root causes of friction in the workplace, the "5 Whys" method was employed. By repeatedly asking "why" in response to a participant's statement, it was possible to narrow down specific behaviours to the underlying emotional and structural motivations.

The interview protocol was designed following an "hourglass" structure to build rapport before addressing sensitive topics. This flow was essential because lactation is an intimate topic, and participants may be prone to social desirability bias if trust is not established first. Three participants were interviewed: two with breastfeeding experience and one who had opted against direct breastfeeding due to a traumatic birth experience. Two of the participants were of Dutch nationality, and the other Romanian. The interviews were conducted over the phone, from which a transcript was produced. All participants signed consent forms prior to the interview and were previously given the information letter to analyse.

3.2.3 Data Analysis

While conducting the interviews did provide an abstract understanding of the problem, in order to solve it, a more detailed approach was taken. More specifically, the thematic analysis method outlined by Braun & Clarke in their 2006 paper titled "Using thematic analysis in psychology" was used. Because the scope of this research was explorative in nature, an inductive thematic analysis approach was taken [7]. This allowed the knowledge to be built bottom-up, using the information given to formulate educated assumptions rather than confirming existing ones. First, questions from answers in the transcripts were separated and Excel sheets were populated with the respective information. This offered the possibility to code the information received based on the more abstract insight it provided. After all answers were coded, the next step was to find the emerging themes between them. For a better visualisation, each code was written onto a sticky note and placed on a whiteboard. When creating the notes, repeating information was merged in order to avoid cluttering the board too much (e.g., "stress as a factor in breastfeeding" and "stress and mental load affecting breastfeeding/pumping" were merged into a "stress" sticky note). This can be observed in Figure 1.

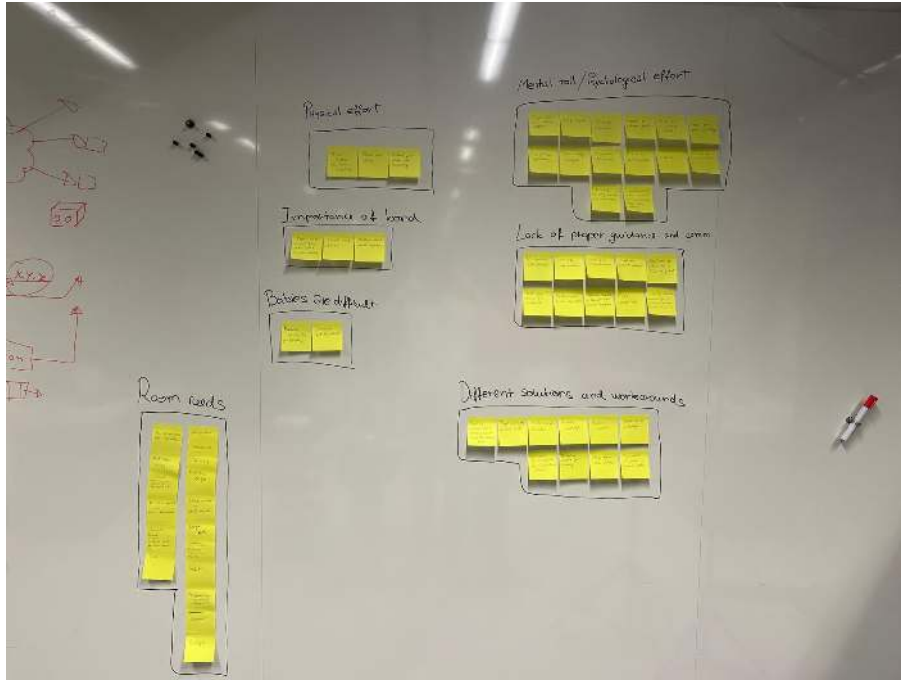


Figure 1: Interview Themes Definition Process

All codes thought to address similar issues were placed together, and, based on the insight provided, the groupings were appropriately named. This process was repeated until no post-its were left unsorted. As a result, seven themes emerged: *Mothers feel discomfort due to physical pain. Stress and uncertainty take a mental toll on the mother's psyche. Breastfeeding is a multi-faceted social dilemma that leads mothers to experience social strain. The bond between a mother and a child is fundamental for breastfeeding. The diverse set of issues caused by breastfeeding leads to a lack of proper guidance and communication. Breastfeeding pushes mothers into purchasing various products, and average rooms are not sufficient for proper care-giving and breastfeeding.*

3.2.4 Findings

In the early stages of subject exploration, it was known that breastfeeding is not easy for either the baby or the mother. However, it was difficult to pinpoint exactly the troubles parents have to go through both physically and mentally during the breastfeeding period. Through the help of the thematic analysis, a deeper understanding of these experiences developed.

Mothers feel discomfort due to physical pain. During breastfeeding, mothers have to get used to bodily functions that were not present previous to the birth of their first child. The interviews have highlighted some of the troubles that come with the ability to lactate, such as blockages of milk ducts, differences in milk flow and breast pains when the mother is not lactating. These identified challenges suggest not only an increase in

physical effort for the mothers but also a high level of adaptability required to understand their bodies and act accordingly. One of the interviewees described their situation as follows: "I tried...to get the baby on the breast, but it was coming out too strongly, so the baby couldn't swallow it." This shone light on a daunting aspect of breastfeeding that was unknown in the beginning: unpredictability.

Stress and uncertainty take a mental toll on the mother's psyche. The mental toll breastfeeding has on mothers has been at the forefront of all the interviews. The most apparent aspect when interpreting the codes was the different levels of psychological effort mothers have to go through and the multitude of faces these difficulties can have. From social pressure to stigma and self-blame, women have to combat expectations coming not only from the outside world but also from themselves. Because so many aspects of breastfeeding can vary from one person to another, more often than not, mothers end up feeling like they are not doing things correctly. This theme was especially apparent in the answers received from the mother who was reluctant to breastfeed: "...the biggest issue I experienced mentally was that it's kind of like a stigma that you should breastfeed a child..." Stress was also identified as a considerable factor in perceived psychological effort: "...I'm having a bit less milk...because sometimes if you start a day and then you're in your head...oh, now I have to pump; oh, now I have to work..."

The bond between a mother and a child is fundamental for breastfeeding. This theme refers to the connection between the mother and her child as a core element of a pleasurable and cooperative experience of feeding. One interviewee highlighted that she realised that breastfeeding is a two-way effort between the mother and the baby, and they need to collaborate: "I can't just expect the baby to do all the work. I also have to help him a little. It's not enough to put the breast in his mouth so he can eat". "I have to do what feels good for me and what gives me the best opportunity to bond well with my child." Moreover, it was mentioned that the entire process of finding the best solutions for both sides is a process consisting of multiple trials and errors, as exemplified here: "I tried baby bottle tips meant for older babies [...]. I tried the older-age tips, thinking the milk would flow faster and help him. [...] But even that didn't work. Only the cheaper bottle worked."

Breastfeeding is a multi-faceted social dilemma that leads mothers to experience social strain. While conducting the interviews, it became evident that some mothers force themselves to behave and act in certain ways in order to conform to "social norms". The self-correction that results from following these unwritten rules may be beneficial in cases where the attitude of the mother is reckless and can damage the well-being of the child. However, in all other situations, they take a toll on the mental health of the

mother by putting pressure on them to make decisions that are "objectively" right, rather than allowing the parent to make choices that are right for them. The following quote from an interview outlines this idea clearly: "...the biggest issue I experienced mentally was the stigma around breastfeeding, the idea that you should breastfeed your child, even if it's not something you feel able or comfortable doing... I wasn't coping well after the birth".

The diverse set of issues caused by breastfeeding leads to a lack of proper guidance and communication. The interviews highlighted this theme with regard to two time dimensions: before and after birth. The need for information is crucial in such a complex process where both one's own body and the life dynamics are changing, and the lack of support and sentiment of an individual struggle outlined by social stigma create feelings of anxiety and uncertainty. Mothers feel a lack of proper instructions on how breastfeeding works in practice, what techniques and products can aid them and their infants during the process, and how unpredictable the encountered challenges can be. Additionally, they highlighted the inefficiency of paediatricians and qualified staff, as well as the lack of feeling of community among the mothers who are facing the same situations. One interviewee mentioned, "And I had to massage the breast and help stimulate lactation somehow, and I didn't know how, because I wasn't properly instructed in the maternity ward," while another highlighted, "I would have liked the nurses who helped me at birth to give more explicit and detailed explanations."

Breastfeeding pushes mothers into purchasing various products. As the market is already extensive, with solutions that support both the infant and the mother, several of these were repeatedly mentioned during the interviews. They range from baby products such as bottles that replicate the mother's breast to solutions meant to aid mothers, such as pumps, protections for leaking or massage techniques. Moreover, substitutes for classical elements of this experience are mentioned as well: bathrooms as substitutes for specialised breastfeeding rooms, formula as a supplement for milk, or thermic bottles as a mode of transport of pumped milk. Some quotes illustrating these alternatives are: "[...] we didn't just stay home. In those cases, the breast pump helped us."; "[...] my baby is quite hungry, so we also supplement with formula milk."; "I tried different methods."

Average rooms are not sufficient for proper care-giving and breastfeeding. Because breastfeeding has so many variables that can change from mother to mother, the recommendations received came from vastly different perspectives. However, an intersection of key features could be outlined. Properties such as privacy, safety, accessibility and comfort were labelled as valuable by all of our interviewees. Approaches on how that can be achieved ranged from the addition of "a little table next to me" to "a larger armchair

or a small couch and a bigger trash bin”.

3.3 Discussion

The interviews address the challenges faced by lactating mothers that extend beyond the need of a dedicated room and a schedule.

A key insight is represented by the stress factor, which is present in all themes. Participants expressed that uncertainty, inadequate facilities, fear of interruption, and pressure to remain productive all contribute to it. This confirms Hentges & Pilot’s (2021) [6] finding that returning to work introduces a “double burden”, where the parent must simultaneously manage professional duties and a physically demanding routine. The interviews reinforced this through personal accounts describing pumping as more of a full-time job on top of regular work.

In the interviews it is shown that rights are often poorly communicated by the staff or unevenly supported, leaving a gap between the policy and the lived experiences. As a result, mothers often navigate through the process alone, relying on the internet or personal initiative. Moreover, the data highlight lactation as an intimate, embodied process that requires psychological safety, dignity, and predictability; these needs are often overlooked in the design of workplace facilities. These insights broaden the scope of the challenge from improving rooms to addressing the wider system of norms, communication practices, and support structures that shape daily experience.

In re-framing the challenge, it was deduced that the core issue is not referring to insufficient lactation spaces only, but rather to a broader systemic issue: the lack of cultural, organisational, and emotional support for lactation within academic workplaces. The limitations of this research include the relatively small sample size and potential positional biases of the design team, who lack personal experience with lactation. However, the consistency of themes across interviews offers a strong foundation for a more holistic design approach.

3.4 Conclusion

The analysis helped clarify the realities lactating employees face and how these experiences affect them. Furthermore, the interviews showed that the challenges they face rarely stem from the physical act of pumping alone. Instead, they come from the workplace culture, unclear communication, emotional strain, and the intimate, embodied nature of lactation.

4 Ideation and Prototyping

4.1 Re-frame

Initially, the challenge of pumping or breastfeeding at work was approached as a primarily logistical and spatial problem, centred on the availability of lactation rooms and compliance with legal requirements. However, insights from the interviews revealed that this framing was incomplete. The challenge can now be reframed as “a systemic issue of emotional, psychological, and organisational support for embodied care work within the workplace.” Breastfeeding at work is not only a physical task; it also involves significant emotional labour, referring to the continuous mental effort required to manage stress, expectations, and social norms while lactating. This is compounded by a lack of psychological safety, meaning that lactating employees often fear interruption, judgement, or negative professional consequences when expressing their needs.

Furthermore, the interviews revealed the inherently unpredictable nature of lactation and infant care conflicts with rigid workplace structures, creating stress and reducing a sense of control. In this context, dignity emerges as a central concern, defined as the ability to perform essential care practices with privacy, comfort, and legitimacy rather than through improvised or marginal spaces. Due to limited institutional communication and guidance, many participants relied on individual workarounds, highlighting a shift of responsibility from the organisation to the individual.

This re-framing shifts the problem from the provision of “bare minimum” facilities toward the design of a supportive system that recognises lactation as legitimate care work and prioritises emotional comfort, predictability, and dignity alongside physical requirements. Overall, the challenge becomes one of organisational and environmental support for lactation at work: where dignity, psychological safety, and predictable access matter as much as the physical room itself.

4.2 Design Vision

Function. The design vision focuses on the aspect of the room itself, more specifically, on creating a room which is supportive both practically and emotionally, as described by the project statement “From Bare Minimum to Emotionally Supportive Lactation Rooms”. To achieve this, a review of the findings uncovers the most important functions for the design.

- The room must allow lactating parents to pump or breastfeed without fear of interruption or breach of privacy.
- The room should enable lactating parents to position themselves comfortably while also being supportive during pumping or breastfeeding.

- The room should be designed to be a calming space, since stress was identified as a central factor influencing both emotional wellbeing and milk production.
- The room’s availability should be easy to access, with a clear form of communication of when it is occupied and for how long, to reduce uncertainty.

Target Group. The target group is lactating parents who use the room to pump or breastfeed during the workday. This group requires a comfortable private space that provides them with the necessities to pump or breastfeed reliably while also being calming and emotionally supportive. Another key characteristic of this group is the need for predictability regarding the occupancy of the room.

An important contextual consideration is that lactating parents may occasionally have young children with them. Although children are not the primary users of the room, their presence can influence the experience of the parent and should therefore be acknowledged within the design context.

Interested Party. The interested parties involved in the design of the lactation room include employers and organisational decision-makers, who are responsible for providing appropriate facilities and ensuring compliance with legal requirements for lactation rooms; the quality of lactation rooms reflects organisational values and impacts employees’ perceptions of workplace support [3]. The responsibility for communicating employee rights, coordinating room access, and managing other organisational practices related to lactation at work falls to the human resources department.

Other interested parties include building managers, as they influence the location and usability of the room, and maintenance staff, who are responsible for its upkeep. In addition, colleagues and supervisors indirectly affect the experience of lactating parents through workplace culture, scheduling practices, and other informal social norms surrounding the use of lactation spaces.

Environment for Use. Lactation rooms are typically located within workplace environments such as universities and offices. In these settings, it is often the case that lactation rooms are situated within multi-purpose spaces, for example, coffee rooms, restrooms, or storage areas. As a result, the lactation room is prone to interruptions due to its shared use. This context introduces practical constraints related to limited space and shared use, which must be taken into account.

Relation with other products. The room should be equipped with facilities that support the needs of the target group, such as breast pumps, milk storage, and a predictable schedule.

Lactation spaces are often shared with other available or improvised workplace spaces, such as coffee rooms or bathrooms, which can create a feeling of vulnerability and a persistent sense of possible interruption, resulting in discomfort and reduced privacy.

Another contextual consideration is that lactating parents may occasionally have young children with them. In these situations, lactation rooms can be more supportive if they include simple features such as toys or other age-appropriate activities to help keep children occupied while the parent is pumping or breastfeeding. Thoughtful additions like these can reduce stress and make the space more practical and comfortable for both parent and child.

The unique quality of the room comes from its design, which embodies the required functionality while also being emotionally supportive.

Distinguishing aspects. The distinguishing aspect of this design vision lies in its focus on the lactation room as an emotionally supportive environment rather than a space that merely meets minimum functional legal requirements.

In addition, the room is envisioned as an integrated part of the workplace rather than a hidden space. By addressing both physical and emotional aspects of lactation, this approach distinguishes itself from conventional lactation rooms that focus primarily on compliance. It instead aims to support dignity and normalisation within everyday work environments.

4.3 Ideation Activities

Paper Prototype 1. The ideation process began with an exploration of existing breastfeeding and lactation room solutions. Inspiration was drawn from professional examples, such as the Healthy Horizons’ portfolio, which offers personalised breastfeeding room setups for organisations [8]. Additional insights were gained from a video titled “Inside a Lactation Room at the U.S. Capitol”, highlighting both strengths and limitations of shared lactation spaces [9]. The lactation room at Stony Brook University further informed the process by showcasing features such as private storage and community-orientated elements like a whiteboard [10]. To broaden the scope beyond Western contexts, spatial and environmental design elements from Osaka Metro lactation rooms were also analysed [11].

Following this research phase, the ideation shifted toward spatial planning. A two-dimensional sketch of the room was created, starting with non-negotiable elements such as the door, sink, and coffee machine, as seen in Figure 2. The lactation chair was then positioned as the central comfort-focused element. Additional furniture components were designed to support modularity and multifunctionality, including an open shelf for pump and personal item storage and a cabinet that integrates a fridge and drawers.

The final stage of ideation focused on experience-enhancing elements aimed at improving both emotional and physical comfort. These included a manual door lock, card-based access to specific features, a tissue dispenser, accessible cleaning supplies, and a sound control unit. The sound control unit, envisioned as a tablet-like interface, allows users

to select or cast preferred music or ambient sounds within the room. To ensure proper maintenance and shared use, elements such as the fridge, drawers, and sound control system were designed to be accessible only via staff card activation.



Figure 2: Paper Prototype 1 - Sketch during ideation process

Paper Prototype 2. The ideation of this prototype was guided by research showing that breastfeeding-friendly spaces need to support not only physical needs but also emotional comfort and a sense of being welcomed. Burton et al. (2024) highlight the importance of comfortable seating, clear information, and clean environments for breastfeeding parents [12]. Using their research as a guiding point, a concept for the room to feel more directed towards lactating parents was developed. The use of the pink colour was chosen to create a calming and caring atmosphere, as studies in colour psychology show that gentle, warm hues can reduce anxiety and promote feelings of safety [13]. Decorative plants were added to make the space feel more alive and relieve stress, drawing on appropriate research [14].

Sketch Prototype 1. The research interviews indicated that the physical environments of the lactation rooms frequently reflect neglect. Participants described these spaces as featuring harsh neon lighting and a cramped, storage-room-like atmosphere. Given that stress is a primary factor that inhibits the physiological let-down reflex, our design vision focuses on transforming these sterile environments into restorative and dignified spaces that prioritise the psychological well-being of the parent. To realise this vision, this prototype showcases the Ambient Personalisation Interface. This system replaces static, clinical walls with integrated LED surfaces capable of displaying immersive natural imagery (such as tranquil forest scenes or rhythmic ocean visuals), designed to support the transition from “worker mode” to “nurturing mode.” The system is controlled using

a wall-mounted tablet interface that allows parents to adjust mood lighting and acoustic masking, including ambient sound or music. By offering a degree of environmental autonomy, the design returns agency to the parent.

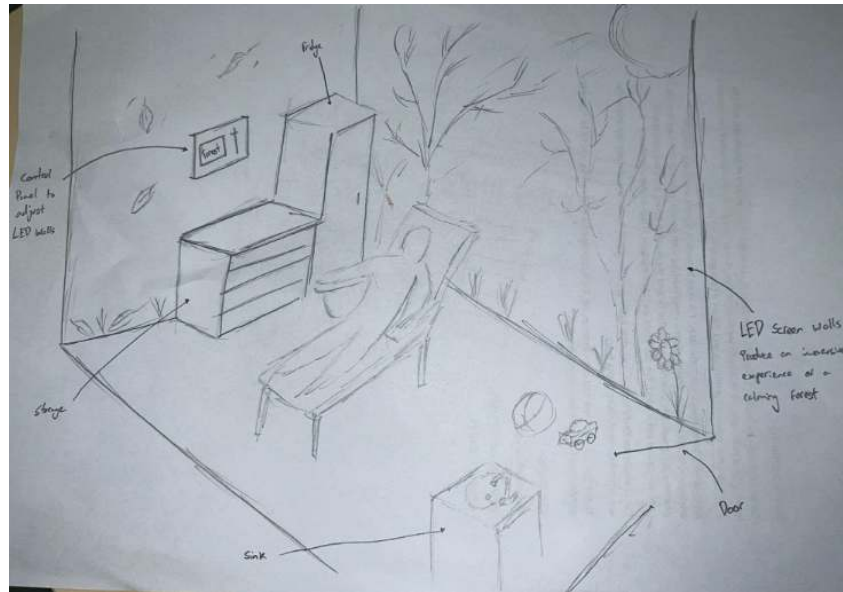


Figure 3: Sketch Prototype 1 - Ideation Process

Sketch Prototype 2. The ideation process began with the exploration of existing workplace spaces and their constraints, such as fixed furniture, room size, and infrastructure. Rather than designing an ideal space from scratch, the focus shifted towards these limitations and working with them to align with the vision of creating a supportive environment.



Figure 4: Sketch Prototype 2 - Inspiration during ideation process

3D Room Model. One topic of interest during the conceptualisation phase of this prototype was immersion. To better grasp the dimensions of objects and the feelings

generated by interior design choices, different methods of visualisation were researched. Looking over industry standards in this field, 3D models emerged as the common solution. This allowed for experimentation with various furniture items from websites hosting 3D assets. Playing around with these models, the idea of using digital twin tools to better map room proportions was developed as a means to support more informed design decisions. As a result, a 3D scan of the existing room was created and projected in the development environment. This established a shared spatial baseline that guided the design decisions that followed.



Figure 5: 3D Model Room - Initial render of the interior

4.4 Scenarios

For this part, the “before” and “after” scenario types were chosen to understand how current solutions work and propose an improved experience. Therefore, the actor, goal and setting are the same for both scenarios, as follows:

- **Actor:** Emily, 32, is a new mother who has returned to work at the DesignLab. She regularly pumps breast milk during the workday to maintain her supply and relieve physical discomfort.
- **Goal:** Emily wants to express milk in a calm, private, and hygienic environment, without feeling stressed, exposed, or rushed.
- **Setting:** A shared breastfeeding room at her workplace, equipped with a chair, table, sink, fridge, coffee machine, and various shared appliances. The room has been used by others before her.

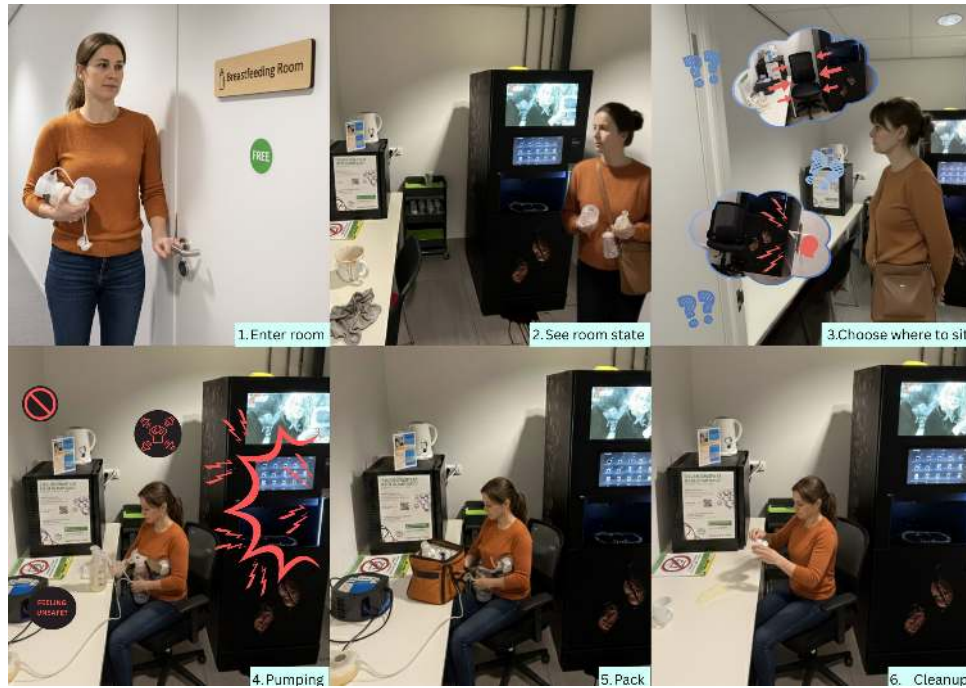


Figure 6: Scenario 1 - Emily Using the Current Breastfeeding Room

1. Enter room: Emily arrives at the breastfeeding room carrying her pump, bottles, and personal bag. She unlocks the door, steps inside, and locks it behind her. At this moment, she expects the room to offer privacy and a sense of calm so she can focus on pumping.
2. See room state: Once inside, Emily pauses to scan the room. She notices leftover cups, surfaces that are not fully clean, and the coffee machine with visible spillages. These details immediately make her more alert and cautious, as she becomes aware that the room is shared and not always reset after use. She then proceeds to clean the table using the napkins from her purse.
3. Choose where to sit: Emily evaluates where to sit before setting up her equipment. One spot is closer to the door and coffee machine, where noise from the hallway and movement feel more present. Another spot feels slightly more secluded but still not fully shielded. She has to choose quickly, balancing comfort, noise, and a sense of safety, even though none of the options fully meet her needs.
4. Pumping: While pumping, Emily becomes increasingly aware of background sounds from the coffee machine and noise from behind the door. The mechanical sounds and the proximity to shared equipment make it harder for her to relax. This tension affects her comfort and sense of privacy during an already intimate and physically sensitive activity, but she continues to pump.
5. Pack: After finishing, Emily carefully packs her pump, bottles, and personal items.

She tries to keep clean and used items separate, working around the limited surface space.

6. Clean-up: Before leaving, Emily wipes down the surface she used. She uses the napkins she has in her purse and some water from the sink. Clean-up feels like a responsibility; she has to improvise rather than follow a process supported by the room.



Figure 7: Scenario 2 - Emily Using the Improved Breastfeeding Room

1. The mother enters the room and locks the door. The indicator outside reassures her that she will not be disturbed, allowing her to relax and focus on herself.
2. She adjusts the room's atmosphere to her needs, setting the light, temperature, sound and visuals to create a calm and safe environment.
3. Before pumping, she cleans the table and her equipment using the materials provided, ensuring a hygienic and reassuring start.
4. She adapts the furniture to her body, finding a comfortable position that supports her during pumping and reduces physical strain. She prepares her materials for pumping.
5. While pumping, she distracts herself with light content, helping the time pass and making the experience feel less clinical and more personal.
6. After finishing, she stores the milk, cleans the space, restores the room to its neutral state, unlocks the door, and leaves, ready for the next user.

4.5 Reflection on ideation

Building on these ideation activities, the process can be reflected upon as a gradual move from abstract ideas toward grounded design decisions. The combination of existing lactation room practices with spatial sketching enabled the transformation of identified user needs into concrete spatial and interaction concepts. The use of different prototype fidelities allowed both functional requirements and emotional qualities to be explored in parallel. This approach helped ensure that design choices remained grounded in empirical insights while progressively refining how care and comfort could be materially and digitally embedded in the room design.

Although breastfeeding and pumping were known to not be instantaneous activities, only after developing the storyboard did the anticipation and preparation required become apparent. Mothers often recognise the need to pump several minutes in advance, which suggests that the room would benefit from a feature that allowed for advance notification and pre-adjustment of environmental conditions, such as temperature, before arrival [15]. This would reduce stress during entry and allow the mother to transition more smoothly into the activity.

The scenario also reveals the importance of transparency upon arrival. To prevent frustration during the process, it should be easily observable in the room if essential materials are still available. This should help mothers decide whether they need to bring additional items. Similarly, the absence of a clearly designated space for personal belongings could complicate the process for the parent. The room is already quite small, meaning there is little area to place items. A reserved corner for bags and personal items would help maintain a sense of order and control.

During the pumping process, similar to what was found during the interviews, waste management emerges as an overlooked aspect of the experience. The storyboard suggests a need for dedicated disposable bags within the room and access to a properly enclosed waste bin, supporting hygienic closure without prolonging time spent in the space. Finally, the visual neutrality of the room contributes little to emotional comfort. Introducing calming or reassuring imagery on the walls could help counteract the clinical feel and support emotional regulation during pumping. This could additionally be used for other purposes as well, allowing for multiple functionalities for the room.

4.6 Prototyping

Differentiating between perspectives offers an opportunity to reflect on the intersection points of the different interpretations. To make use of the diverse set of identities in the team and to assure consideration of all approaches, members have proceeded to create five different prototypes, one for each person. These are presented in the form of interior design ideas, which encapsulate the different features found necessary during the

understanding and ideation process.

4.6.1 Prototyping Activities

Paper Prototype 1. This prototype focuses on the redesign of elements and furniture in the proposed breastfeeding room of DesignLab. To maintain the multifunctionality of the room for building management purposes, the coffee vending machine was kept in place.

First, the plastic desk chair is replaced with a dedicated nursery chair, more comfortable and with an incorporated footrest. It can also provide additional comfort features such as blankets or pillows for back support.

Next, between the chair and the vending machine, closest to the position of the mother, there is an open shelf cabinet to support the pump. Designed with openings big enough for a maternity bag, it provides the space for additional personal items. On the other side of the chair, there is the main furniture piece: a wooden desk cabinet with an incorporated fridge and drawers dedicated for three (or more) types of objects: cleaning supplies, entertainment for the mother, and toys (for the situation where the room is used for breastfeeding, not only pumping). On top of this cabinet, there is a dedicated screen to allow the parent to choose the music or sounds they want. Between the armchair and this furniture element, there is a wall-mounted paper tissue dispenser.

The ideation process directly sustained the design choices by connecting them to existing practices and also the pain points of the current solution. The research highlighted that emotional and physical comfort are usually lacking in current usual solutions even though they are crucial in the process of lactating. Consequently, the redesign of existing elements in the room prioritised these aspects: the chair was changed from a desk chair to a dedicated armchair, and the supporting desk and mug cart were replaced with secure and purposeful cabinets.

Moreover, the security features, such as card-based access, were chosen to aid the privacy and safety of the user and their belongings. As the space is shared, the access levels can increase the safety of the storage of the pumped milk and assure the environment has controlled sound, further balancing user comfort with shared-space management considerations.

Overall, the ideation process ensured that each design choice was not only functional but also experience-driven, focusing on reducing stress, increasing autonomy, and supporting the emotional well-being of the room's users.



Figure 8: Paper Prototype 1

Paper Prototype 2. To reduce uncertainty and empower users, a brochure provides clear information on breastfeeding, responding to the lack of guidance many parents experience. A chair designed specifically for lactation supports physical comfort, while antibacterial toys ensure that children are safely entertained, without compromising hygiene. Finally, a tablet for booking appointments gives parents more control over their time and privacy, helping to reduce stress around room accessibility by allowing users to schedule their next appointment.

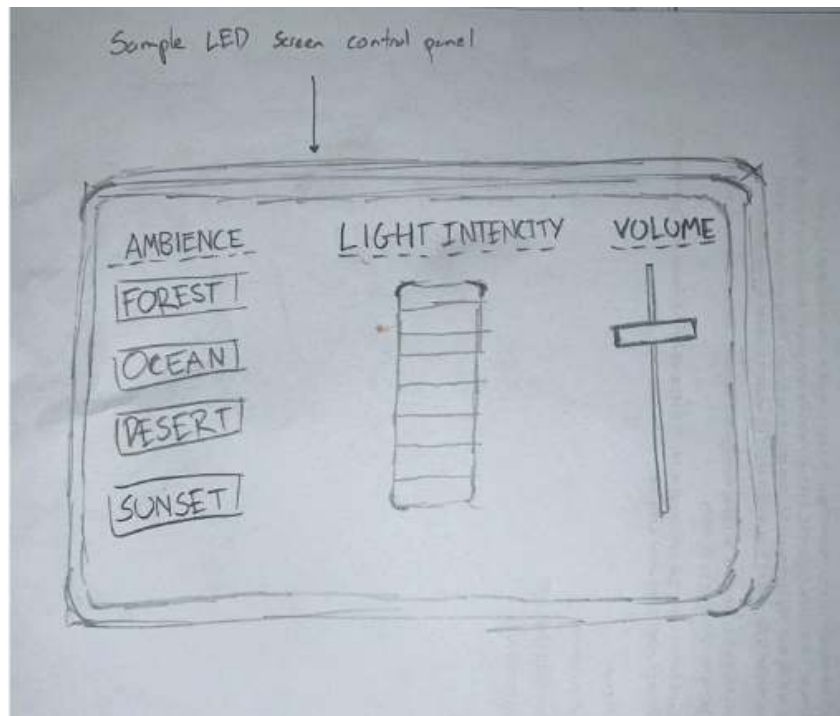


Figure 10: Sketch Prototype 1

Sketch Prototype 2. Design decisions prioritised essential elements that support both physical comfort and emotional ease. A comfortable chair, accessible storage, and a compact refrigerator were prioritised, while unnecessary or overwhelming elements were removed or downsized. The alignment of the existing table with the cupboards and the sink was retained to ensure feasibility and continuity with the space. Cupboards play an important role in reducing visual clutter and maintaining a calm environment. They are used to store essential hygiene and care items, such as baby wipes, surface towels, and cleaning supplies for breastfeeding equipment, elements that are currently missing but crucial for a supportive space. Dustbins and cleaning supplies are kept hidden yet accessible from a height, preventing strong odours while remaining within easy reach. The fridge for milk storage is placed within a closed cabinet to maintain hygiene and privacy, with careful consideration of height to avoid excessive bending. The room also includes a clearly visible first aid kit to ensure quick access in emergency situations. In addition, a storage drawer for over-the-counter medicines allows the space to double as a mental health or mindfulness room. The existing table is retained as a flexible surface for breastfeeding kits or personal items. Adding access to power outlets enables charging devices or optional work or personal use, depending on the parent's preference. Because the room also functions as a prayer space, the design supports multi-purpose use. A soft carpet and small shoe rack improve comfort for prayer while also creating a clean play area for an accompanying child. Large, unnecessary objects such as the oversized coffee machine are replaced with a compact tea and coffee setup. Cupboards store cups and cutlery, and small food items such as protein bars or cookies further enhance comfort. To

reduce stigma around breastfeeding, the purpose of the room is kept transparent. Books and magazines for parents, along with toys for children, are placed on open shelves or behind glass cabinets. Some sections of the cabinet serve as lockable storage for parents' personal feeding supplies. A key addition is a wall-mounted tablet that allows users to control lighting, music, and room access. This enables the room to adapt to different needs, such as breastfeeding, prayer, mental rest, or childcare, strengthening user agency and emotional comfort.



Figure 11: Sketch Prototype 2

3D Room Model. A 3D environment offered the opportunity to play around with different materials, objects, and layouts. This method of prototyping enabled the possibility of deeper immersion in the environment and the ability to see the room and interaction directly from the point of view of the mother using the camera feature of Blender. By doing this, elements of the room that were first apparent upon entering it were identified, along with the affordances created by the space and layout.

During this process, it was observed that if a smart-room touchscreen interface were to be implemented, it would be preferable for it to be positioned close to the table where the mother remains for most of the time, rather than near the entrance. Although positioning the original light switches and the smart screen in similar locations may appear logical from a functional perspective, consideration of the added option for experience customisation indicates that placing the screen in the area where the mother is most comfortable and likely to reconsider settings is the more appropriate choice.

In addition, a cleaner approach to item storage was incorporated into the environment. The currently implemented metal tray was found to be prone to leaving coffee marks on the floor and to contribute to a sense of neglect within the room. As an alternative, a wooden cabinet was introduced, with shelves designated for mug storage and drawers

allocated for item storage. This arrangement ensures that items are not openly visible to all users, while each drawer provides a secure space for depositing materials needed for pumping or breastfeeding. Furthermore, the existing chair was replaced with a more comfortable, compact chair that can be adjusted in both angle and height to suit the mother's needs. Similarly, a sit-stand desk was introduced to allow adjustment of the table height according to user preference. Finally, the fridge was positioned on the cabinet to free up table space and provide additional room for placing bags and clothing.

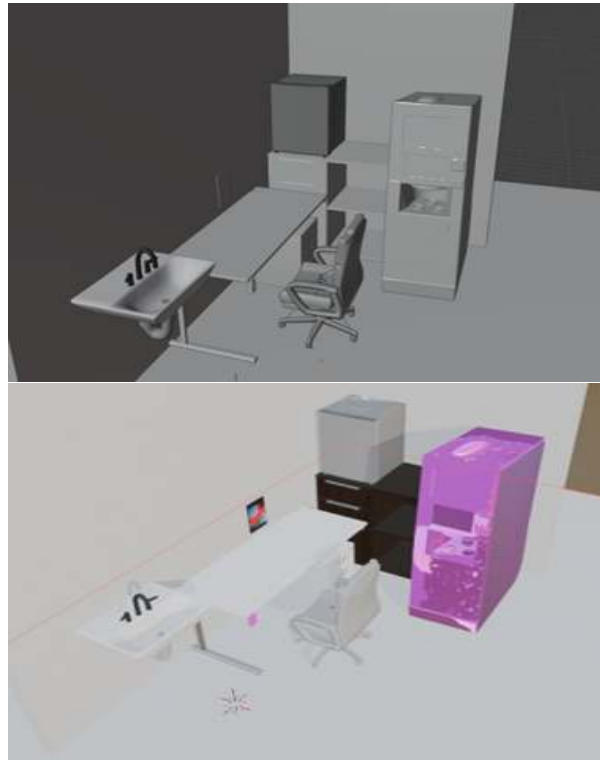


Figure 12: Blender 3D Environment of the Improved Room

4.6.2 Final Guidelines

The prototypes developed explore multiple possible approaches to the interior design of the lactation room in the Design Lab. Across these iterations, a range of features and design choices have been examined and systematically justified. The subsequent step in the development process involves consolidating these insights into a clear and definitive set of proposed features and spatial changes. This set of guidelines can be understood as the primary design artifact emerging from the preceding phases. In addition, it serves as both a recommendation for relevant stakeholders and as a proposed response to the challenges mothers currently face in the existing room. These requirements are presented in Table 1.

ID	Requirement	Description	Design Rationale
R1	Communicative lock system	Lock automatically indicates occupancy outside and confirms privacy inside.	Reduces fear of interruption and increases psychological safety.
R2	Dedicated lactation chair	Chair designed specifically for lactation use.	Addresses physical discomfort during prolonged pumping or breastfeeding.
R3	Arm and foot support	Chair includes arm and foot support.	Supports relaxed posture and circulation.
R4	Centralised layout	Chair, table, storage, and controls positioned within arm’s reach.	Minimises movement and cognitive load during use.
R5	Integrated hygiene and storage	Closed storage, milk fridge, and cleaning supplies provided in-room, positioned within comfortable reach from a seated position.	Prevents improvisation and supports dignified milk handling.
R6	Discreet waste handling	Enclosed waste solution integrated into the space.	Maintains hygiene without visual or sensory disturbance.
R7	Ambience control system	Preset-based control of lighting, sound, and visuals.	Reduces stress while avoiding cognitive overload.
R8	Temperature control	User-adjustable or preset room temperature.	Supports physical comfort and relaxation.
R9	Seating-adjacent controls	Control interface positioned near the chair.	Enables mid-session adjustments without disruption.
R10	Room reservation system	Advance reservation or clear scheduling of use.	Improves predictability and reduces anticipatory stress.

Table 1: Core design requirements for an emotionally supportive lactation room.

4.6.3 Moving Forward

So far in the ideation and prototyping process, the team has focused on developing interior design ideas and room structures that are context-aware and try to reduce the difficulties mothers encounter while trying to breastfeed at work. This objective has fol-

lowed the outlined design vision and the framing of the problem, that being the pivot from bare minimum to emotionally supportive lactation rooms. The team understands the commitment that would be necessary to develop, test and implement such a large number of features and changes to the room. Therefore, moving forward, the team will focus on only one identified feature. The development and evaluation of this characteristic should not be interpreted as sufficient to alleviate the exhaustion associated with breastfeeding at work. It is rather a symbol and a first stepping stone to proving the interior design choices have been outlined eloquently and, therefore, encourage interested parties to move away from the bare minimum, with the help of the presented findings.

The focus in the next chapters is going to be put on the third bullet point outlined in the function section of the design vision, which states, "The room should be designed to be a calming space, since stress was identified as a central factor influencing both emotional wellbeing and milk production."

Motivation for this first comes from overlapping the prototype ideas and identifying the common importance set on room ambience. Team members collectively agreed that the existing room feels too empty and impersonal for the scope of breastfeeding. While the room is required to be multi-functional, as outlined in the design vision, it should nevertheless provide an appropriate level of comfort. Another reason for this choice is that it results in the development of a prototype that allows for the exploration of human-technology relations. This creates an opportunity to identify the methods in which technology can be applied in this context through evaluation.

Therefore, the focus will be on developing a lo-fi prototype of a smart-room interface, the system responsible for creating the desired atmosphere of the space. The features and interactions are explored in the following chapters.

4.6.4 Storyboard

The proposed system is intended to function as a tool for effectively shaping the ambience of a room. Drawing on environmental psychology research by Mehrabian and Russell (1974), which demonstrates that emotional responses to environments emerge from the combined influence of multiple sensory stimuli, three key functionalities have been identified [16]:

1. The system needs to be capable of adjusting the colour and intensity of the lighting within the room.
2. The system needs to be able to present comforting or mood-specific visual content.
3. The system needs to be able to play atmosphere-appropriate sounds or music.

Apart from these three, with consideration to the requirements identified previously, three more features emerge as important:

1. The system should support scheduling and reservation of the room to ensure predictable access.
2. The system should allow users to adjust or pre-set the room temperature to maintain physical comfort.
3. The system should support account-based access management through university card authorisation.

However, initial development focuses on the first three affordances, while the remaining three are considered for subsequent implementation phases. The result is the following storyboard that describes the expected user-system interaction.



Figure 13: Storyboard of the user-system interaction

1. The mother enters the lactation room, establishing the context of use and the transition from a shared work environment into a private space intended for breastfeeding or milk expression.
2. The room is characterised by harsh overhead neon lighting and a predominantly office-like atmosphere, which disrupts comfort and reinforces a clinical or work-orientated setting rather than a supportive and calming environment.
3. A tablet mounted on the wall becomes visible, presenting an interface with options to adjust lighting, visual content, and ambient sound, signalling the availability of environmental control.

4. The interface is used to browse through a set of predefined ambiance presets, allowing the user to quickly explore different combinations without requiring extensive interaction or configuration.
5. Upon selecting a preset, the room's lighting, visuals, and soundscape change simultaneously, resulting in a noticeable shift in ambiance from an office-like environment to a more calming and supportive setting.
6. Within the adapted environment, the mother can express milk in a relaxed and comfortable manner, indicating a reduction in environmental stressors and an improved overall experience.

4.6.5 Low-fi Prototype

For the low-fi prototype of the final design, the focus was on simulating the experience of interacting with the proposed system. First, the main components of the final proposed design were chosen:

- **Input Screen:** the main element of interaction with the system; this is where the user can change the environment either in predefined themes or personalise the experience.
- **Displays:** the output displays where predefined pictures are shown according to the theme; can only be turned on/off, no other interaction is possible.
- **Sound system:** plays different sounds according to preference or can be connected with own music applications for more customised experience; can only be interacted with through the Input screen;
- **Light ambiance:** multiple light sources to create the desired atmosphere; similar to the sound system, it can be accessed only through the interface.

Components. After defining the elements, the next step is to identify ways of low-fi prototyping for each of them to make sure the interaction and functionality are easy to understand and comfortable for the user. Consequently, each element was designed with at least two states: Idle and Active. For the Interface, Display and light ambiance, further states were needed to showcase the interaction more effectively.

Interface. For the Interface element, only 3 states were deemed necessary to explain the functionality, as seen in Figure 14.

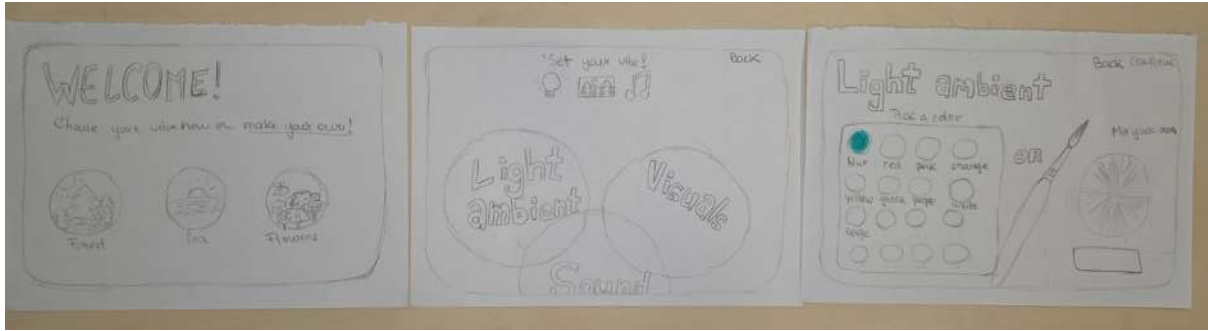


Figure 14: Interface States

The first screen, representing the Idle state or, in other words, the Welcome screen, showcases the message first seen by the user when the interaction with the system starts. It also contains two main options, "Choose a vibe" or "Make your own", and three predefined choices of themes: "Forest", "Sea" and "Flowers". Next, the second screen can be accessed through the "Make your own" button and represents the main menu of the customisation feature. It contains the three categories of elements that make up the system: Light ambiance, Visual, and Sound. By clicking any of these, the respective menu will be shown. An example consists of the last screen state, the Light Ambiance interface. Here, the user can either choose a predefined colour or select a specific one from the colour wheel.

Display. The displays, which will be wall-mounted throughout the visual space of the user, showcase different collections of images and animations predefined by category or theme. Initially, the screens are empty until the user chooses either a preset or category of images from the main interface. Once the choice is made, they display the respective collection. These changes in state can be seen in Figure 15.



Figure 15: Display States

Sound System. This component contains a surround sound system to create a good sound balance in the room. It only plays the audio cues selected directly by the user or the respective tunes of a certain preset. In this prototype, there were used only two papers shaped like speakers, as seen in Figure 16.



Figure 16: Sound System States

Light *ambiance* In reality, the light ambiance will consist of LED strips and RGB lamps, but for prototyping purposes, the choice was made only to showcase the result of the interaction; therefore, only two states are present: White (Idle), which is the default room light, and Colour, in this case green, which is the hue chosen by the user, as seen in Figure 17.

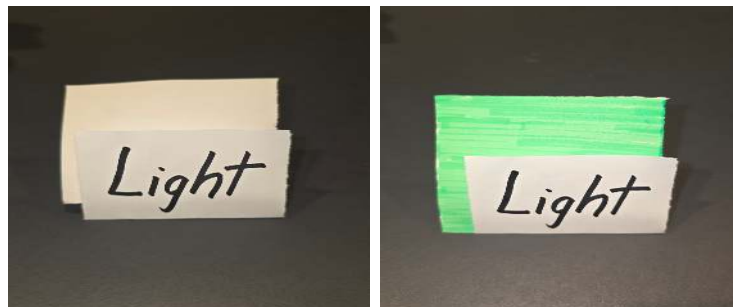


Figure 17: Light States

System States. After the components of the system and their states were defined, the system prototype was put together to showcase the main interaction flow and its consequences on each element.

Defaults. The default settings are composed of the idle mode state of the components (Figures 14 to 17) which come together in a full system as in Figure 18 (left).



Figure 18: Main Default State: Idle mode

Choices and Interaction. From the Default state, there are two main choices to be made by the user: either choose an ambient preset or customise the environment. Therefore, we distinguish two main cases: Case 1: Customisation (Figure 19) and Case 2: Presets (Figure 20).

Case 1: Customisation. Here we refer only to Figure 19. This case starts from clicking "Make your own" on the Welcome screen. It covers the three main ambience elements: Light, Visuals and Sound. From here, the user can choose exactly which aspect they want to customise, while the others remain in Idle state until manually changed. After selecting one of the options, a dedicated menu appears, as seen in Figure 19 (right). This example covers the Light element: the user can either choose from a series of predefined colours or choose a specific one from the Colour wheel.

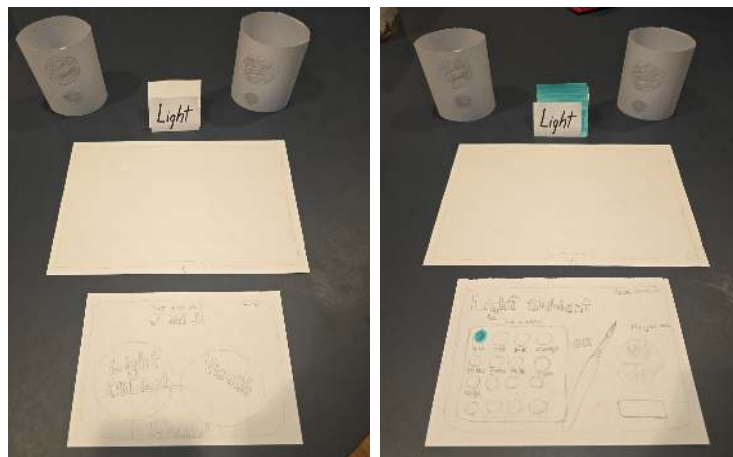


Figure 19: Interactions Choices Customization: Choice "Make your own" (left), Choice "Light Ambient" (right)

Case 2: Presets. Here we refer only to Figure 20. This case starts when making a choice between the multiple presets on the Welcome screen. On the paper prototype, there

are currently three main choices. The user clicks on the respective choice as exemplified in the figure. One such interaction then changes the ambience of the system: for the "Forest" choice, the lights turn green, the display shows a natural forest landscape, and play bird chirping and rustling of leaves. Similarly, the "Sea" preset changes the environment to a more marine, beach-like ambience.

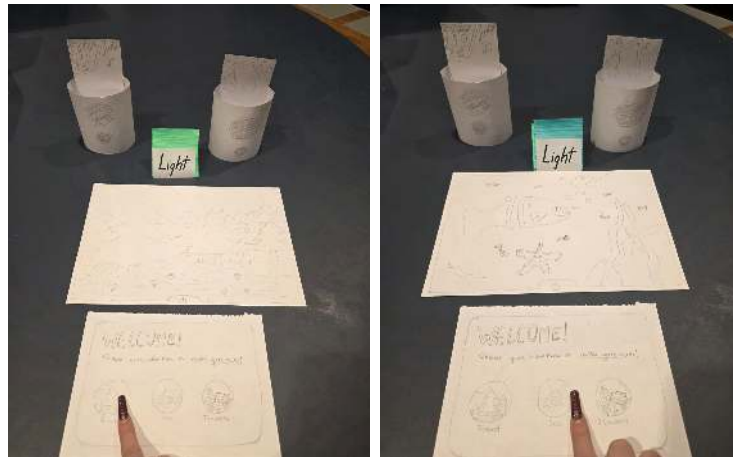


Figure 20: Interactions Choices Presets: Choice "Forest" preset (left) and Choice "Sea" preset (right)

4.7 Reflection on Ideation and Prototyping Process

Defining the design vision more clearly allowed for a smooth transition into the ideation stage. The ability to trace back to the guiding principles and the initial research phase helped substantially in keeping all ideas aligned with the intended objectives and to further map them to real-world challenges encountered by mothers. Looking back on this process, such an approach helped keep the project on track and ensured that perspectives, even though different at times, aimed for the same outcome.

During the prototyping phase, these diverse viewpoints were used to explore different interior designs. Moving beyond traditional paper prototypes allowed for the design problem to be envisioned through multiple lenses. Developing and comparing the arguments behind design choices of each prototype provided a deeper understanding of the various directions the project could take. Differentiating between these choices served as a reminder of the complexity involved in providing an effective solution, even within a limited scope such as interior design. Visualising these design options made it possible to outline a definitive list of recommended room changes.

This process was used as a tool to identify areas of intersection that would result in the discovery of a tangible product, as required by the assignment. A meaningful outcome of this phase was the transition from a broad exploration of the interior toward a specific solution that could be effectively tested in the evaluation chapter. The initial approach to the problem was general, a decision that has led to the development of prototypes that

integrate a larger range of features. While this allowed for an in-depth exploration of many possibilities, it eventually needed to be concretised into a testable product. The "Moving Forward" subsection describes this transition. To avoid the potential misunderstanding of this choice and to re-articulate the overall objective, it was emphasised there that the development and implementation of this product do not represent the full extent of the proposed solution but that it rather represents a component of a broader set of features envisioned for the room. These features are outlined in the aforementioned subsection and are further reflected upon in the discussion chapter.

Ultimately, this phase highlighted the importance of well-defined design guidelines and the value of a thorough understanding of the context, particularly in a process that places the user at the foundation and builds upon that understanding at every stage.

4.8 Conclusion

This chapter set out to explore how the findings from the understanding phase could be transformed into design direction through the ideation and prototyping techniques. This process was shaped by initially re-framing the challenge from a solely logistical and functional problem to an issue of emotional, psychological and organisational support, which further moved the focus toward the design of an emotionally supportive lactation environment.

Throughout ideation activities, scenario development and multiple prototyping techniques, this chapter highlights how abstract, subjective findings, such as discomfort, stress and mother-child bond, can be further materialised into room layouts, elements of furniture, concepts and design requirements recommendations. Paper sketches and prototypes, scenarios and 3D renders aid both phases and provide the opportunity to explore the design directions from different viewpoints: functionality, feasibility and experiential quality. The ideation and initial prototyping phases resulted in a first design artifact: a clear set of design guidelines for lactation spaces.

A main takeaway from this phase is that emotionally supportive lactation rooms cannot be achieved through isolated features but instead require a system which includes spatial layouts, furniture, sanitary products, ambience and environmental control. From here, the focus was narrowed from a broad room layout and feature redesign to a specific emphasis on ambience, which represents the balance point between initial research, design vision, ideas and the testing feasibility within the scope of this project.

In the next phase, the project will build upon these insights by further developing and testing one central feature: the ambience smart-room system. This evaluation will provide empirical grounding for the design assumptions established in this chapter and inform future iterations of the proposed system.

5 Evaluation

5.1 Experiential Prototype

5.1.1 Elements

The conceptualisation of the next experiential prototype started from the paper prototype presented previously. Next, each component was matched with the easiest functional alternative available, as follows:

- **Tablet Interface:** the main component for controlling the whole system. The mock-up interface was done using Figma. In the system prototype, this mock-up will be run on an iPad to showcase the control unit. It allows users to view and manage settings related to the components of the system: ambient lighting, screens, speakers and the room temperature. By gathering all controls in one place, the interface reduces complexity and makes the experience more intuitive. Model screens can be seen in Figure 21.

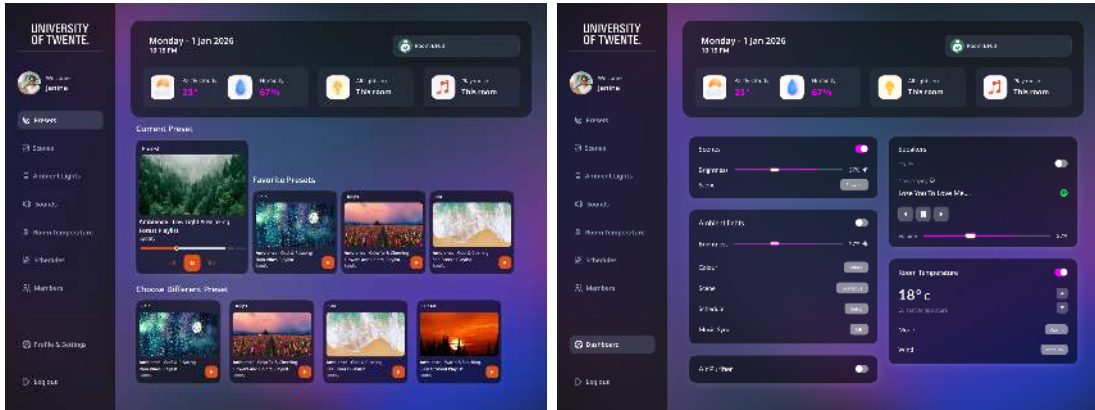


Figure 21: Mock-up Screens for Control Interface

- **Visual Displays:** wall-mounted output screens that show ambient visuals and interface feedback. The visual displays support the emotional aspect of the experience. They present calming visuals that correspond with the selected ambience and reinforce the overall atmosphere of the room.
- **Sound System:** a JBL speaker that plays ambient sounds or music chosen by the user. All sound and music interaction is managed through the tablet interface.
- **Light Ambiance:** An RGB lamp with adjustable brightness and colour in order to shape the mood of the space; similar to the sound system, lighting presets and hues are chosen only through the interface.

5.1.2 Element Setup in Evaluation Room

Once the components were defined, the experiential prototype brought them together into a single coherent system. The Figma prototype simulates how the system reacts to real-time human input by changing the lighting, visuals, and sound. In this way, the interface becomes the control unit of the overall experience, guiding the system based on user input. Then, the fitting space for the testing of the system was deemed to be the very main room discussed in this project: the lactation room situated in Design Lab. Therefore, the following placement of elements was made with regard to the current available space and layout of the room:

- **Interface.** This component is placed on the desk for easy accessibility.
- **Displays.** The components are placed in such a manner that the peripheral vision of the user is covered by the theme visuals. Rather than functioning as interactive screens, they act as output elements that visually communicate the mood chosen by the user.
- **Sound System.** The speaker, which is placed on the desk in order to offer a surround sound, completes the multisensory setup by adding an auditory cue to the space. Through the interface, users can select ambient sounds or music that help mask external noise and create a more private and relaxing environment.
- **Light Ambiance.** The RGB lamp, which is placed on the ground, offers the user the light ambiance, allowing it to change the colour and the intensity of the light. This allows the room to transform from an office room to a softer and more comforting one, supporting emotional regulation during lactation.

To better visualise these placements, pictures with the setup can be seen in Figure 22 and Figure 23.¹



Figure 22: Testing setup

¹There were no photos taken to the participants due to privacy. The person in the pictures is a member of the team.



Figure 23: Testing setup presets

5.2 Evaluation Scope and Method

5.2.1 Method

To evaluate the lo-fi prototype, the team decided to employ the use of the cooperative evaluation method; this participant-centred usability method positions users as active collaborators in the process of assessing the prototype, rather than being treated as passive subjects [17]. This method encourages participants to openly share their thoughts, interpretations, and emotional responses live while they are interacting with the prototype.

The selection of the cooperative evaluation method is closely aligned with the goals of the project itself. Since the design of the prototype aims to support psychological comfort, it is essential to learn not just what users do, but why they respond in the ways that they do. It is this point which makes cooperative evaluation the right fit, as it enables participants to articulate how the prototype makes them feel, giving the team direct insight into the user experience, which is particularly valuable when evaluating experiential and emotionally sensitive designs [18]. These insights will aid in addressing one of the main issues the team is attempting to solve, which is facilitating lactating parents to feel calm enough to transition from a work mindset to a calm nurturing one.

This method is also especially appropriate given that it is being applied to a low-fidelity prototype, meaning that at this stage of the process the goal is not to assess technical or visual polish, but to identify conceptual strengths, usability concerns, and experiential gaps early in the design process [19]. The “think-aloud” nature of cooperative evaluation allows participants to co-reflect with the design team; this offers unique insights grounded in lived experience that may not emerge through pure observation or quantitative metrics alone, as would be the case with other evaluation methods.

From a practical standpoint, cooperative evaluation is also an efficient formative method, as it allows the team to collect a large amount of rich qualitative data within a single session [17]. Unlike quantitative approaches that focus on task completion, this method provides depth by revealing how users perceive autonomy, comfort, and cognitive load within the prototype. This makes it particularly well suited for identifying early

usability challenges and refining design decisions while there is still scope for meaningful iteration.

5.2.2 Scope

The goal of this evaluation is to examine how the prototype supports the psychological wellbeing of lactating parents within a workplace lactation room. In particular, the evaluation focuses on whether the combined use of lighting, sound, and visual content results in an altered ambient experience that aids in reducing stress and, in turn, supports the lactating parent in transitioning to a calm state.

In addition to this, the team seeks to gain insight into the participants' experience with the prototype interface, specifically whether the design decision to provide a more limited degree of environmental autonomy through a simplified user interface helps to promote a smoother and less cognitively demanding transition from a work mindset to a calm, nurturing one.

Building on insights from the research phase, this evaluation responds directly to the identified “double burden” experienced by lactating employees, in which professional responsibilities must be managed alongside physically and emotionally demanding care work. Rather than measuring task efficiency or performance, the evaluation adopts a formative and experiential focus, aiming to understand users' subjective perceptions of comfort, calmness, and cognitive effort when interacting with the system.

5.2.3 Participant tasks

To conduct an accurate evaluation, participants will be required to complete a set of tasks intended to represent the everyday use of the prototype. While completing these tasks, participants will be encouraged to speak aloud as part of the cooperative evaluation process [17]. During this process, the team will observe and note any issues encountered with the system. The tasks chosen are:

1. Select a room ambience using one of the predefined presets.
2. Adjust the lighting settings within the selected preset.
3. Adjust the sound settings within the selected preset.
4. Adjust the visual scene within the selected preset.
5. Schedule a booking for the lactation room using the interface.
6. Adjust the room temperature.

5.2.4 Survey

After the session has been conducted, participants will be issued a survey to complete. The purpose of this survey is to gain further insight into participants' psychological experience with the prototype. The survey is designed using a mixed-methods approach; the first section contains Likert-scale items, followed by a small number of open-ended questions [18].

The Likert-scale questions are framed to capture information regarding participants' perceived comfort, autonomy, cognitive load, and atmospheric change. The Likert scale was chosen as it helps identify patterns across participants and provides a systematic approach for measuring subjective experience. The open-ended questions allow participants to elaborate, reflect, and identify issues they may have experienced during the interaction. This qualitative data provides a level of depth and context that the Likert scale alone cannot offer and aids in explaining why participants responded as they did to the scaled items.

Overall, the combination of cooperative evaluation and a post-session survey enables methodological triangulation, strengthening the evaluation by integrating real-time experiential data with reflective self-report measures [19].

5.2.5 Research Question

Given the exploratory nature of the prototype and the use of cooperative evaluation, the study is guided by one overarching research question:

How does the prototype influence lactating parents' perceived psychological comfort and sense of transition from "worker mode" to "nurturing mode" within a workplace lactation room?

To further operationalise this research question, the evaluation will focus on the following analytical aspects during data analysis:

- Participants' perceptions of the atmospheric shift from a clinical, office-like environment to a more restorative setting.
- The extent to which interaction with the interface is experienced as supportive or cognitively demanding, particularly under conditions of time pressure or emotional vulnerability.
- Users' interpretations of environmental control in relation to autonomy, dignity, and emotional ease.

Together, these aspects provide a structured lens through which the experiential and interactional qualities of the prototype can be examined.

5.2.6 Participants

All participants were students at the University of Twente, aged between 21 and 25. None were lactating parents; all participated as empathetic proxy users, asked to imagine themselves using the room as a lactating parent under mild time pressure. Participants varied in prior familiarity with the room and levels of technical confidence. A more in-depth overview for each participant’s demographics can be found in Table 2.

5.3 Evaluation Data

The use of the **think-aloud method** allowed the evaluation to gather a large number of feedback comments that could be qualitatively analysed. The notes written during the user testing were collected in one place so that common themes could be identified using a thematic analysis informed by prior analytical categories. Similarly to what was done during the understanding process, information was simplified and evaluated on the subject it tackled. As a result of this, the following themes emerged:

- Presets as Cognitive Anchors Under Low Mental Capacity
- Minimalist Adjustment as a Strategy for Maintaining Calm
- Sound as Boundary Rather Than Atmosphere
- Scenes as Non-Demanding Backgrounds
- Scheduling as a Source of Psychological Safety and Legitimacy
- Comfort Features Interpreted as Signs of Care

Furthermore, qualitative insights were triangulated with quantitative results obtained through the **final survey**. Having empirical evidence for the observed themes increased confidence in the results and shone light on further improvements. Usability and overall opinion on the system were numerically graded using the answers gathered at this step.

5.4 Evaluation Findings

5.4.1 Think-Aloud

When observing the notes from the user tests, a specific approach to the interface emerged. Testees were considerably more focused on the comfort aspect of the system rather than its intricate features. More specifically, participants interacted with the system just enough to make the room more calming, without extensively exploring the multiple customization opportunities offered to them. Moreover, the changes they indeed make usually consisted of small, minimal alterations, avoiding considerable shifts in the environment. Comments

such as "If I'm stressed, I wouldn't want to think too much." and "As long as it stays subtle, it's fine" meant that the interface should, before anything else, provide the user with a quick and simple option to make the room more calm. This approach is further explored by analysing the six identified themes.

Presets as Cognitive Anchors Under Low Mental Capacity. Presets were included in the experiential prototype with the scope of providing easy and direct access to the main atmospheric capabilities. During the evaluation, this feature achieved its proposed goal. Multiple users acknowledged the help provided by the pictures of the presets. One participant mentioned, "I'm mostly choosing based on how it looks." Moreover, it was discovered that a large set of presets would not resemble a priority in the future, since users were consistently looking to make simple decisions. Rather than exploring all available options, participants repeatedly selected a calm or nature-based preset and adjusted minimally afterwards. For future work, the changes the presets are about to make should be clearly communicated to the user. The comment of one participant brought awareness to this problem: "I'm trying to understand what exactly this changes."

Minimalist Adjustment as a Strategy for Maintaining Calm. As mentioned previously, it was observed that customisation of the atmosphere was kept to a minimum. Interaction was characterised by testing once, confirming comfort, and stopping further engagement. Continued interaction itself was perceived as potentially disruptive, rather than beneficial. For example, P1 reverted quickly after testing colour, with the comment: "I don't want to change too much."

Sound as Boundary Rather Than Atmosphere. Sound was consistently treated as a functional boundary to block external noise and create privacy, rather than as an immersive or expressive feature. Participants prioritised control, muting, and predictability over richness or variety. The initial design choice to include audio in the system was meant as a tool to develop an atmosphere. However, it was used multiple times with the intention of isolating the room from the outside world. This aligns with the findings gathered in the understanding process, where external noise was identified as a stress inducing factor when breastfeeding. P6 expressed anxiety about unpredictability: "Unexpected sound would stress me out."

Scenes as Non-Demanding Backgrounds. Visual scenes were generally engaged with briefly and then left unchanged. Participants valued scenes that did not demand attention, framing them as atmospheric rather than interactive elements. Nature-based or non-campus imagery was particularly effective in helping participants mentally disengage from institutional or work-related contexts. Comments made by both P1 and P6 encapsulate the essence of this theme: "It's nice if it's just there in the background" and "It shouldn't demand attention." Ultimately, it was concluded that scenes function best when they remain peripheral, supporting mood regulation without introducing additional stimuli that compete for attention.

Scheduling as a Source of Psychological Safety and Legitimacy. The scheduling feature elicited some of the strongest emotional responses. Participants interpreted it not merely as a functional booking system but as a mechanism that legitimised their presence and reduced social anxiety. P4 described it as emotional permission: “This feels like permission to take this time.”

Comfort Features Interpreted as Signs of Care. Temperature adjustments were minor but symbolically important. Even when the interaction was minimal, participants interpreted the temperature control as an indication that their bodily comfort had been considered. As one participant put it, “Someone actually thought about comfort.” Therefore, even if not fully implemented, the feature showed the user that a concern has been put on comfort during development. This reiterates the many ways the HCD framework can impact final impressions on design.

5.4.2 Post-test survey

The survey results suggested high levels of agreement with statements related to perceived control over the environment. On the one hand, the visual and sensory rating scales showed a consistent shift towards more relaxing and dignified perceptions of the space after the interaction; however, these results also showcase that a high level of customisation tends to become unimportant when fast atmosphere change is desired. This can be seen in the results presented in Figure 24.

2. Psychological Impact & Experience

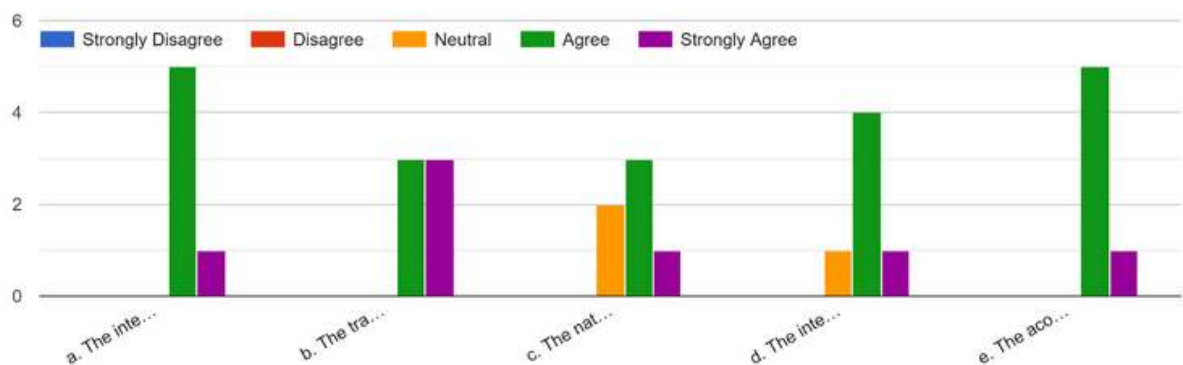


Figure 24: Opinions from the survey on control of the room (a), change in atmosphere (b), nature presets (c), intuitiveness (d) and acoustic masking (e).

Changes in perspective in the room can be observed in Figure 25. The system did a good job at providing a more relaxing, comfortable, and dignified space. However, the addition of screens, interfaces, lights, and an audio system could have had an impact on the perceived cramped-ness.

These quantitative trends reinforced the qualitative findings by demonstrating that the identified themes were reflected not only in the verbalised experiences of the participants but also in their overall evaluative judgements.

3. Visual/Sensory Perception: Evaluate the room atmosphere after using the interface.

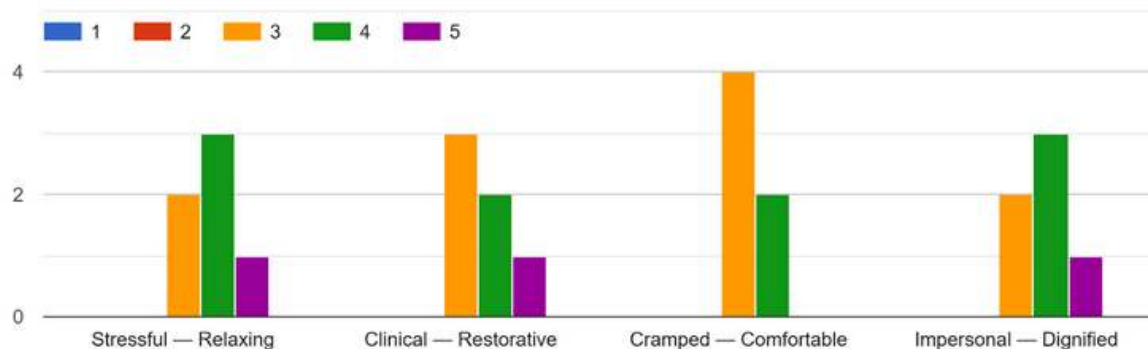


Figure 25: Results from the survey on changes in perspective in the room.

5.5 Discussion

The evaluation indicates that the prototype is experienced as most supportive when it enables a fast, low-effort change in atmosphere. Participants consistently relied on presets and made only small adjustments, suggesting that extensive customisation is less relevant in moments of low mental capacity. In this sense, the interface primarily functions as a quick way to reduce cognitive demand while still offering a sense of control.

Furthermore, the modalities were often interpreted differently than originally intended. Sound was mainly treated as a boundary to mask external noise and create privacy, while visual scenes were preferred when they remained subtle and non-demanding in the background. Together, these findings suggest that multisensory elements are most effective when they support calm without requiring continuous attention or interaction.

Finally, some features appeared to influence psychological comfort beyond direct use. Scheduling was strongly associated with legitimacy and reduced social pressure, while temperature control was seen as an indication that bodily comfort had been considered. At the same time, survey responses suggest that adding multiple components may affect the perceived spaciousness of the room, indicating that experiential improvements must be considered alongside spatial constraints.

5.6 Conclusion

This evaluation chapter presented the deployment of the experiential prototype in order to examine how the proposed system supports psychological comfort within a workplace

lactation room. The findings indicate that the prototype was effective in shaping the perceived atmosphere of the space while also revealing important constraints that inform future iterations. Participants consistently favoured quick and effortless interaction, often relying on predefined presets rather than exploring detailed customisation. This behaviour reinforces the assumption that, in emotionally demanding contexts, reducing cognitive effort is more valuable than offering extensive control.

The multisensory components of the system were primarily interpreted as environmental regulators rather than interactive features. Sound was frequently used to mask external noise and enhance privacy, while visual content and temperature adjustment contributed to a sense of separation from the surrounding workplace. Together, these elements supported immersion and reduced anxiety, suggesting that subtle environmental changes can meaningfully influence perceived calmness without requiring sustained attention. In addition, the scheduling feature played a significant role in legitimising users' presence in the room, helping to alleviate social pressure and uncertainty.

The post-test survey further supported these observations. Responses indicated that relaxation and comfort were prioritised over fine-grained control, with presets and ambient sound having a stronger impact than manual adjustments. Overall, the combination of cooperative evaluation and a post-session survey enabled methodological triangulation, strengthening confidence in the evaluation outcomes while maintaining a formative focus appropriate to the prototype's level of fidelity.

6 Overall Discussion

This project was undertaken for the Human-Centered Design course, which explicitly addresses challenges related to diversity, equity, and inclusion, with a particular focus on health and well-being. This course revisits the design process, not as a purely technical activity, but as a reflective and iterative method in which understanding people, power relations, and context is central.

The design challenge addressed in this project concerns supporting lactating individuals or parents who may feed their babies at their workplace. Although legislation in the Netherlands mandates the provision of lactation rooms and breastfeeding breaks [4], research showed that access to suitable spaces remains inconsistent and that existing rooms often meet only minimal functional requirements. This gap between legal standards and lived experience formed the starting point of the project. Rather than assuming that compliance with guidelines equates to adequate support, the project investigated how lactation rooms are experienced in practice and what qualities meaningfully affect their use.

Building on the course structure of understand, ideate, and evaluate, with continuous reflection and reframing across phases, this chapter now discusses how the project's

understanding of the problem evolved. Rather than focusing on the final design outcome, the discussion reflects on how insights from user research, contextual analysis, and theory informed design decisions. Attention is given to how design qualities such as functionality, comfort, support, and emotional experience were carefully defined and operationalised within the real constraints of corporate workplace settings. Maintaining a critical perspective, the discussion moves beyond the design outcome, identifying the methodological and contextual limitations of the study, such as the reliance on proxy users and lab-based testing. Findings are situated within the practical realities of design research. Acknowledging these constraints is essential for defining the project's scope and provides the necessary foundation for the final recommendations, ensuring that future steps toward holistic workplace lactation support are both grounded and realistic.

6.1 Reflection on the Design Process

The design process had to be grounded in a human-centred approach that prioritised understanding lived experience within real organisational contexts. At the outset, the project was motivated by a dissatisfaction with how workplace lactation rooms are typically treated as compliance-driven facilities rather than intentional spaces for care. Early assumptions framed the problem as one of insufficient amenities or poor spatial quality. However, insights from user interviews and contextual inquiry challenged this framing by revealing that the primary difficulties associated with workplace lactation were not only spatial but also temporal, emotional, organisational, and sometimes societal.

Participants described lactation at work as an activity performed under time pressure, often fragmented by meetings, deadlines, and spatial availability. This highlighted that the lactation room cannot be understood in isolation; it is embedded within a broader workplace system that shapes how, when, and with what mental state the space is used. In line with interaction design theory, this reinforced the view that user experience emerges from the relationship between people, artefacts, and context rather than from the artefact alone [20].

As a result, the notion of functionality was reconsidered. Within this project, functionality does not refer simply to whether a room contains required equipment but to whether it reliably enables lactation within existing workplace constraints. A functional space, in this sense, supports the activity without demanding additional planning, negotiation, or cognitive effort from the user. This distinction was important in avoiding a purely technical interpretation of design quality. [21]

Similarly, comfort was not treated as an aesthetic or sensory quality alone. Instead, comfort was understood as a reduction of physical strain and mental tension during use, particularly given the limited time available to lactating parents. This interpretation was directly informed by user accounts and reflects research emphasising that design for

wellbeing must be grounded in situational realities rather than idealised scenarios [22].

6.2 Reflection on Design Decisions

This section reflects on the key design decisions made throughout the project and the reasoning behind them. These decisions were not treated as isolated choices but as responses to specific insights about user experience, workplace context, and cognitive and emotional constraints identified during the research phase.

One initial central design decision was to prioritise ambient conditions over the addition of interactive features. This decision emerged from interview findings indicating that lactation at work can be stressful due to time pressure and cognitive fragmentation. Adding complexity, even with supportive intent, risked increasing user effort rather than reducing it. Drawing on interaction design principles, this decision reflects an understanding that usability and experience are shaped not by feature quantity but by how well a system aligns with the user’s situational capacity [20].

A second key decision concerned the deliberate limitation of user choices. Rather than offering extensive customisation, the design implemented predefined settings. This was informed by a clarified definition of autonomy as the ability to achieve a desired state with minimum effort, rather than as maximum control. This interpretation was grounded in user accounts describing mental fatigue and the need for predictability during lactation sessions. From a design justice perspective, this decision also aimed to avoid shifting the burden of adaptation onto users already navigating structural constraints [2].

Decisions related to spatial layout and control placement were similarly guided by many considerations. Controls were positioned to be accessible from a seated position, reflecting the physical demands of lactation and the limited mobility during use. This decision operationalised the project’s definition of support as environmental anticipation of needs, rather than reactive assistance.

Overall, these design decisions demonstrate how reframing the problem space led to guidelines for improving the room to support the parents. Eventually, the care activity directly influenced what was designed: a controllable room for comfort.

6.3 Limitations

The results of the evaluation provided valuable insights into the “desirability of environmental autonomy”. However, there were some limitations to the evaluation that reduce its overall generalisability and ecological validity.

The primary limitation of the evaluation lies with the demographic of the participants in the study. As noted in Design Justice [2], the most valuable ingredient in equitable design is the “full inclusion of, accountability to, and control by people with direct lived experience”. Due to constraints, the participants consist of university students, none of

whom are currently lactating or pumping. The data gathered therefore only represents psychological comfort rather than physiological validation. While the participants confirmed that the ambient experience felt calming, they were not able to verify whether the experience can successfully trigger the “let-down reflex” or if the interaction with the system significantly added to the cognitive load experience for the intended user.

During testing the team employed the use of the “wizard of Oz” approach in order to simulate the smart room’s features. While this lo-fi prototype experience is effective for exploring the design space, this method introduces some constraints and inconsistencies. Due to human error and the prototype not being a complete functioning system. The experience, orchestrated by the team, is not as seamless as intended, resulting in there being some technical latency. Furthermore, as Benyon [20] notes in *Designing Interactive Systems*, prototyping functionality can sometimes diverge from the actual embodied experience. Because participants interacted with a tablet while “imagining” the room changing around them, we cannot fully predict if the “double burden” of managing a digital interface while assembling pumping equipment would increase stress in a real-world scenario. A subsequent HCD cycle would require a high-fidelity integration within a physical mock-up room to test this “interaction flow” more accurately.

The evaluation was conducted in a controlled environment which does not account for real-world stressors. In a workday context, lactating parents often have to face time pressure and “anticipatory stress” regarding their return to work duties. The protocol used in the evaluation had users complete tasks in a relaxed setting, resulting in the study lacking ecological validity. It is possible that features deemed “delightful” in a calm test (such as browsing ambience presets) might become “obstacles” when a user has only a 15-minute window to pump.

Finally, the team acknowledges our positionality as students without the lived experience of lactation. Despite our efforts to centre an “ethics of care”, our design decisions, such as prioritising visual “immersion”, may reflect our own biases toward technological solutions. We must consider if a non-technological solution (such as improved soundproofing or a more ergonomic physical chair) might have addressed the “structural domain” issues more effectively than the high-tech intervention we prioritised. This reflection highlights the need for reflexivity in all future design stages to ensure we are not over-complicating a sensitive human experience with unnecessary technology.

6.4 Social implications of the design

When it comes to the social implications, three main issues can be distinguished, as follows: the normalisation of lactation within workplace environments, a shift of responsibility towards institutions rather than individuals, and the need to actively manage organisational culture and perceived fairness among colleagues.

Firstly, our environmental guidelines combined with the tested prototype of the ambience control system highlight a shift from a bare-minimum, hidden lactation room to a first step towards an emotionally supportive dedicated space. As the design is intended to enhance the existing room in question, the multifunctionality of the space was preserved throughout the ideation process to ensure that the final design remains a visible and legitimate part of daily working life, while still allowing for privacy during milk expression. This moves the focus towards the consideration of pumping and breastfeeding at the workplace as a normal part of working life rather than something that needs to be concealed or treated as conspicuous. However, evidence suggests that workplace support is mediated not only by the physical room itself but also by co-worker attitudes, perceived stigma, and perceptions of fairness surrounding time and space accommodations. [1, 23, 24]

Next, another key implication follows from normalisation within the community and concerns the shift of responsibility: rather than lactating parents improvising solutions, the organisation itself must provide adequate options and clear procedures for its employees. This aligns with evidence that lactation room quality and organisational support shape satisfaction, perceived ease, and perceived support for expressing milk at work, including support from supervisors and co-workers. [3] Moreover, experimental evidence suggests that higher-quality lactation rooms can reduce anticipated or experienced stress and increase perceived organisational support, indicating that design quality is not merely aesthetic but consequential for well-being and the feasibility of combining breastfeeding and work. [25]

Finally, introducing more visible and supported lactation spaces can surface tensions around equity and resource allocation, particularly when colleagues perceive accommodations as costly or disruptive. Prior work shows that perceived personal cost and beliefs that breastfeeding is a private issue can reduce perceived fairness of workplace accommodation. [24] This makes it important that the organisation communicates the rationale for lactation support, formalises access and frames the space as part of inclusive workplace practice rather than an informal favour. [23, 26]

6.5 Moral implications of the design

When it comes to the moral implications, three main issues can be distinguished, as follows: the safeguarding of privacy and dignity, the protection of autonomy through non-intrusive interaction, and the governance of power and fairness around access, data, and control.

Firstly, while the proposed ambience control system is intended to support comfort and emotional regulation, any form of digital layer in an intimate setting raises questions of privacy beyond mere physical. Even when no explicit sensing is implemented, fea-

tures such as scheduling, authentication, or logs of room usage may create the perception of being monitored, which can undermine psychological safety and the core aim of the space. A useful framing is Nissenbaum’s notion of contextual integrity, which suggests that privacy is maintained when information flows remain appropriate to the norms of a specific context; in this case, the lactation room context carries strong expectations of confidentiality and minimal traceability. [27] Consequently, the moral priority is to minimise collected data, limit retention, and avoid visibility of identifiable usage information to third parties, ensuring that organisational support does not inadvertently become a mechanism of oversight. [26, 28]

Secondly, the system must support user autonomy in moments that are often time-sensitive and emotionally loaded, without turning support into additional labour. In practice, this means that the interface and interaction design should reduce cognitive load, favouring quick presets and non-demanding control rather than complex choices. This aligns with work on implicit interaction, where good systems are designed to be helpful without persistently requiring attention or explicit negotiation, particularly in contexts where users are already managing competing demands. [29] From a value-orientated perspective, Value Sensitive Design is useful here as it frames autonomy, dignity, and well-being as design core values, encouraging designers to translate moral aims into concrete interaction constraints rather than treating ethics as a post-hoc consideration. [28]

Finally, moral implications extend beyond individual use to questions of power: who gets to decide what the room is for, who controls the settings, and whose needs take priority when the space is multi-functional and shared? A design that is experienced as supportive by lactating parents can still be ethically fragile if access is governed informally, if users must negotiate permissions, or if room improvements primarily serve institutions rather than human needs. Design justice and literature offer language for these concerns, emphasising accountability, the distribution of benefits and burdens, and the need to centre impacted stakeholders in decisions about governance and infrastructure. [2, 30] This makes it important that the further implementation is accompanied by transparent norms and by continued engagement with non-proxy users to avoid embedding external assumptions into an emotionally sensitive space. [2]

6.6 Future work

Building on the insights gained from the evaluation conducted as a part of this initial Human-Centred Design cycle, future work will focus on transitioning the prototype from a low-fidelity lab-based concept to a high-fidelity socio-technical system. To do this the following three key areas have been identified as priorities for the next iteration, each of which targets a limitation discussed earlier.

Firstly, to address the “lived experience gap”, the next phase must move beyond

just proxy users. Following the principles of Design Justice [2], the team intends to conduct Participatory Design workshops with lactating parents across diverse workplace sectors. Conducting studies with participants of this demographic would ensure that the ambient experience is not just “calming” in the general sense but specifically tuned to the physiological and emotional needs of those actively breastfeeding. It will also enable the team to sufficiently investigate if the system is working as intended by triggering the “let-down reflex”. Additionally, conducting new studies with the intended user group will potentially uncover new insights about other limitations or issues that would need to be addressed.

The next technical iteration will replace the current “Wizard of Oz” setup with a fully integrated high-fidelity prototype. As Benyon [20] emphasises, the “embodied experience” is critical to interaction design, so future work will involve the designing of this high-fidelity prototype. The system will have a fully functional interface, and its features will operate as intended, being that the lights, ambience, and visuals will work together fluently, and the parts used will be of a higher standard, contributing to a more immersive experience. This will allow the team to test the “Double Burden” hypothesis: observing whether a lactating parent can intuitively manage the interface while simultaneously handling physical pumping equipment or breastfeeding needs without an increase in cognitive load.

To validate the design’s impact on long-term well-being, the project must move into real-world contexts. Future evaluations should consist of longitudinal field studies within an actual office environment. This would allow the team to gather in situ data, facilitating the evaluation of how the system performs under genuine time pressures of the working day and whether the ambient experience is truly capable of having a “restorative” effect on the user, and if so, whether this would continue after multiple uses. Again, this brings the prototype system closer to real-world applications, stress-testing it to discover where it could be further improved.

Finally, future work should explore a more holistic approach to the lactation room. While ambient personalisation addresses psychological comfort, a complete HCD solution must integrate physical requirements. To do this, the team aims to explore how the digital interface could coordinate with physical furniture, such as adjustable ergonomic seating or smart storage for milk, creating a truly unified “smart room”. There is also room to explore the integration of the room scheduling system that we theorised as a part of the design vision, with the potential of integrating a system where workers swiping cards would grant specific features if they are a lactating parent, and adding another layer of privacy and security.

7 Conclusions

As a conclusion, this study began by addressing the gap between lactating employees' practical and emotional needs and the facilities or organisational practices intended to support them. After identifying the core issues, namely inadequate lactation spaces alongside limited organisational support, the project was able to define a clearer scope and objective. By focusing on how lactation is experienced in practice, the interview phase helped refine both the framing of the challenge and the direction of the design work. This allowed the team to move from an organisational and functional view of the problem towards one grounded in emotional support, inclusion and psychological safety. The reframing then guided the design vision, where experiences such as stress, discomfort and the need for privacy were translated into design directions that prioritised predictability, dignity and a sense of safety.

Building on this foundation, the project progressed through ideation and prototyping activities that supported the exploration of interaction and atmosphere within the room. Through sketches, renderings, models and paper prototypes, scenarios were created to better understand the user journey and the points where support is most needed. This process resulted in the development of a deliverable kit that includes an interface alongside ambience elements, intended to demonstrate how environmental control can shift the room experience. The artefact was evaluated using a combination of cooperative evaluation and a post-test survey, producing qualitative insights alongside structured self-report measures.

Overall, the study provides empirical grounding for the role of environmental autonomy in perceived comfort within an emotionally sensitive workplace context. At the same time, the outcome should be understood as a first step rather than a complete solution. The prototype does not represent the full scope of what is required for holistic lactation support, yet it functions as a test ground that helps validate key assumptions and offers a useful basis for future development within real-world constraints.

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A Participant Profiles

Table 2: Participant profile overview

ID	Age	Gender	Programme	Level	Tech	Room familiarity
P1	22	Female	Psychology	BSc	Low–Medium	Never used
P2	24	Female	Interaction Technology	MSc	High	Once/twice
P3	21	Male	Technical Computer Science	BSc	High	Never used
P4	25	Female	Psychology	MSc	Low–Medium	Once/twice
P5	23	Female	International Business Administration	BSc	Medium	Regular use
P6	24	Male	Computer Science	MSc	High	Once/twice

B Think-Aloud Notes

Table 3: Task-based think-aloud notes and quotes per participant Part 1

ID	Task 1: Preset	Task 2: Lighting	Task 3: Sound	Task 4: Scene	Task 5: Schedule	Task 6: Temperature
P1	Focused on imagery rather than names; hesitated; selected the calm nature preset. <i>Quote:</i> 'I'm mostly choosing based on how it looks.'	Small brightness changes; briefly tested colour, then reverted to neutral. <i>Quote:</i> 'I don't want to change too much.'	Looked first for mute/on-off; set low volume. <i>Quote:</i> 'I just want it to block other noises.'	Treated visuals as background; minimal interaction after first selection. <i>Quote:</i> 'It's nice if it's just there in the background.'	More deliberate; double-checked times before confirming. <i>Quote:</i> 'Knowing this is booked would already make me less stressed.'	Minimal adjustment; increased temperature once, then stopped. <i>Quote:</i> 'Warmer feels more comforting.'
P2	Scanned and compared presets quickly; used presets as a baseline. <i>Quote:</i> 'This feels like a quick-start option.'	Tested brightness range (high to low), then settled on low-medium; colour optional. <i>Quote:</i> 'Brightness already does most of the work.'	Checked full control first; toggled mute; made a small volume adjustment; avoided Spotify. <i>Quote:</i> 'I want to know I can turn it off instantly.'	Compared scenes quickly; adjusted brightness slightly; stopped interaction. <i>Quote:</i> 'I'd probably choose one and leave it.'	Completed booking quickly. Used plus/minus controls; set once, ended interaction. <i>Quote:</i> 'I'd set this once and forget it.'	Increased and decreased temperature <i>Quote:</i> 'This is smooth.'
P3	Initially searched for direct settings. <i>Quote:</i> 'If I'm stressed, I wouldn't want to think too much.'	Tested high/low extremes; returned to mid-low. <i>Quote:</i> 'Middle ground feels safest.'	Prioritised mute; kept low volume. <i>Quote:</i> 'Sound is mostly about making the room feel closed off'	Brief interaction; selected one scene; no customisation. <i>Quote:</i> 'I don't think I'd need this.'	Strong engagement; completed booking; paused briefly after. <i>Quote:</i> 'This prevents awkward situations.'	Minimal interaction; slight increase; would not adjust repeatedly. <i>Quote:</i> 'I'd set this once and not touch it again.'

Table 4: Task-based think-aloud notes and quotes per participant Part 2

ID	Task 1: Preset	Task 2: Lighting	Task 3: Sound	Task 4: Scene	Task 5: Schedule	Task 6: Temperature
P4	Read headings/labels; scanned imagery slowly; chose calm nature preset. <i>Quote:</i> 'It feels like the room is being prepared for me.'	Lowered brightness gradually; selected warm tone; avoided extremes. <i>Quote:</i> 'Bright light makes me feel exposed.'	Avoided silence; set low background sound. <i>Quote:</i> 'A bit of sound makes it easier to relax.'	Engaged longer; selected forest scene; adjusted brightness down. <i>Quote:</i> 'It slows me down a bit.'	Serious/deliberate booking; paused after completing. <i>Quote:</i> 'This feels like permission to take this time.'	Minor increase for bodily comfort. <i>Quote:</i> 'If I'm cold, I can't relax.'
P5	Compared to coffee-room familiarity; selected calm preset quickly. <i>Quote:</i> "This already doesn't feel like the same room."	Lowered brightness to shift social feel; avoided colour settings. <i>Quote:</i> "Lower light feels more private."	Adjusted volume intuitively; used sound to reduce awkwardness. <i>Quote:</i> "It makes it less awkward to be alone."	Strong reaction to non-campus imagery; pause after selection. <i>Quote:</i> "It doesn't feel like campus anymore."	Viewed as boundary-setting; prevents misuse; clear purpose. <i>Quote:</i> "This is what would stop people from misusing the room."	Brief interaction; slight increase; interpreted as care signal. <i>Quote:</i> "Someone actually thought about comfort."
P6	Analytical approach; looked for what parameters are affected; treated preset as starting state. <i>Quote:</i> "I'm trying to understand what exactly this changes."	Assessed control structure; small brightness changes; wanted defaults/reset; avoided colour. <i>Quote:</i> "I don't want to guess what the previous user did."	Tested mute immediately; treated sound as risk unless controllable. <i>Quote:</i> "Unexpected sound would stress me out."	Minimal interaction; accepted only subtle/peripheral scenes. <i>Quote:</i> "It shouldn't demand attention."	Evaluated in terms of enforcement; concern about false confidence. <i>Quote:</i> "Otherwise it gives false confidence."	Preferred precise numeric control; used plus/minus over dial. <i>Quote:</i> "I trust numbers more than visuals."

C Post-Test Questionnaire Responses

Table 5: Post-test evaluation responses per participant

Question	P1	P2	P3	P4	P5	P6
The interface allowed me to feel in control of my environment.	Agree	Strongly agree	Agree	Strongly agree	Agree	Neutral
The transition from clinical white to warm hues made the space feel less like a storage room.	Agree	Agree	Neutral	Strongly agree	Strongly agree	Agree
The nature presets helped me disconnect from work responsibilities and transition to a nurturing mode.	Agree	Neutral	Neutral	Agree	Strongly agree	Neutral
The interaction felt intuitive and did not add to my cognitive load.	Strongly agree	Agree	Agree	Agree	Strongly agree	Neutral
The acoustic masking provided a sense of privacy and psychological safety.	Agree	Strongly agree	Agree	Agree	Strongly agree	Agree
Stressful – Relaxing	4	5	4	5	5	4
Clinical – Restorative	4	4	3	5	5	4
Cramped – Comfortable	4	4	3	4	5	4
Impersonal – Dignified	4	5	4	5	5	4

Table 6: Open-ended reflections per participant

Question	P1	P2	P3	P4	P5	P6
How did the changes in lighting and sound influence your emotional state?	Felt calmer and less rushed	Helped shift focus away from work	Subtle but noticeable calming effect	Made the room feel more personal	Strong sense of relaxation	Reduced distraction but remained alert
Was there any part of the interaction that felt like a distraction or added to your cognitive load?	No	Slightly during first use	No	No	No	Sound controls felt risky
Suggestions for improvement	Clearer default states	Faster preset access	Fewer options	None	Temperature memory	Numeric controls